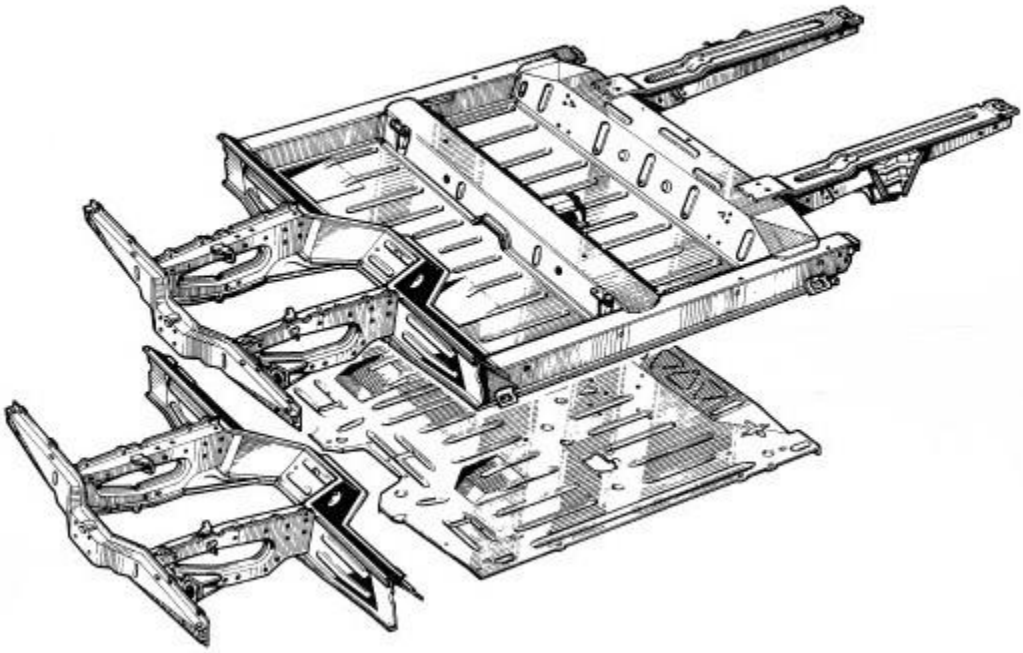
Designing Vehicle Structures



[1]

Renault 4 chassis simulation and validation

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Introduction

The goal of this project is to get acquainted with accurately simulating vehicle structures. To obtain realistic results, the simulation must represent reality as closely as possible. A validation test can be performed to validate the computed output and verify the resemblance of the simulation.

This assignment includes modeling of vehicle structures in computer aided design (CAD) software, determination of realistic loads, creation of a finite element model (FEM), improving the FEM for reduced computation time, simulation of a specific load case, and validation of the simulation.

This project was sponsored by R-Quatre.nl, providing a brand-new Renault 4 chassis for the validation test.

The Renault 4

An economy car built by the French automotive company Renault between 1961 and 1994, the Renault 4 or R4 was a great commercial success, of which eventually over 8 million units where produced.

Founded in 2019 at the HAN University of Applied Sciences, the HAN_4L rally team has since participated multiple times with various models of the Renault 4 in the *4L Trophy Rally* in the Moroccan Sahara Desert. Since, the team has thrived and expanded, generating academic attention for the Renault 4.



Figure 1: A Renault 4 (HAN_4L, 2020), the vehicle model of which the chassis is used in this project.

Platform chassis

"A platform chassis is a separate chassis for a car or small vehicle. It is distinguished from other forms of chassis by being arranged as largely a single flat steel sheet, usually with additional box section stiffeners." [2]

The Renault 4 is one of the best examples incorporating a so-called platform chassis. As can be seen in the picture below, two lateral box sections from the main longitudinal rigidity of the structure. There are three lateral box sections in the main structure of the chassis; the front triangular structure that attaches the chassis members to which the front suspension and engine is mounted, a central lateral beam on which the driver and passenger seats are positioned, and lastly the rear lateral section that is simultaneously the attachment point for the two longitudinal beams supporting the trunk area. All the mentioned members are attached to the floor panel, an essential part for the rigidity of the assembly.

One of the oddities of this French designed little vehicle, is the difference in left and right wheelbase. Because the vehicle is equipped with a peculiar double torsion bar rear axle design, the attachment point of the right axle pivoting support is positioned a bit further towards the rear. This asymmetric difference of about 48mm can clearly be seen in Figure 8 in the following chapter explaining the SolidWorks model of the Renault 4 chassis.

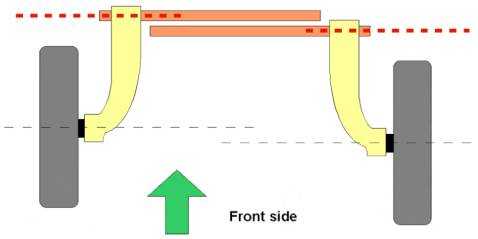


Figure 2: Schematic representation of the double torsion bar rear axle. [4]



Figure 3: Picture from a Renault 4 advertisement depicting the chassis and the seat positions. [3]

SolidWorks model

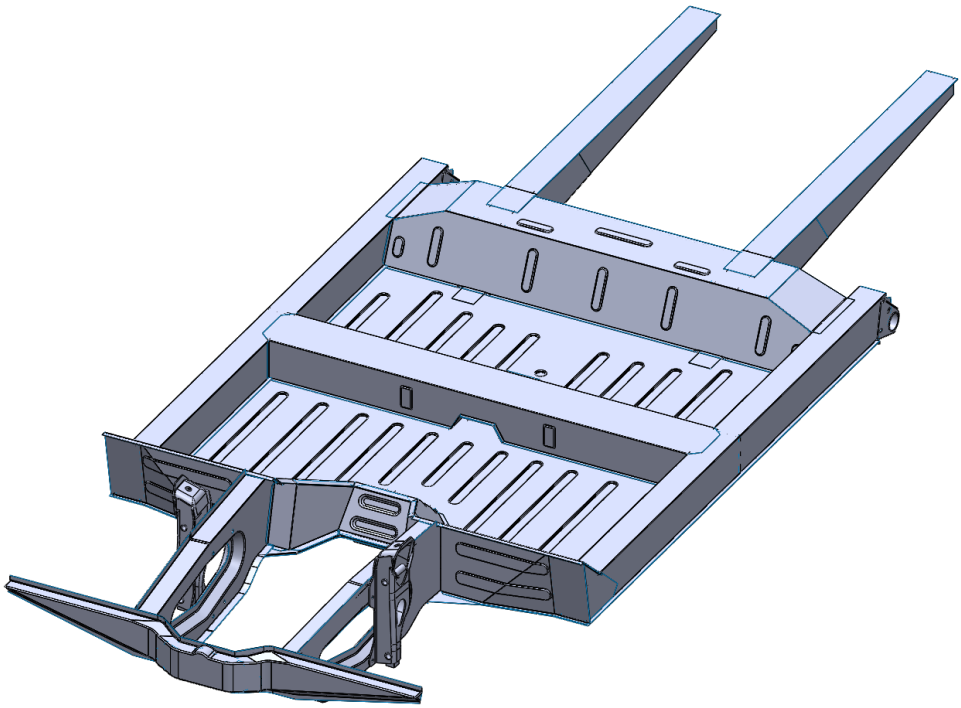


Figure 4: Isometric view of the chassis model in SolidWorks.

Similarly to reality, the model of the chassis is built up from multiple sub parts. To prevent having to manually update each part after a dimensional alteration in a single part, all parts' dimensions are based on a shared 'master sketch'. The advantage of this approach is ease of design and reduced design time. Furthermore, updating basic dimensions within the master sketch are updated in each part. Although this saves a lot of time and reduces the chances for mistakes, each time the master sketch is updated when also the chassis assembly file is opened, a considerable waiting time is to be expected as the master sketch underneath each individual part needs to be updated.

The parts for the main assembly are mostly created using sheet metal feature of SolidWorks, the resulting assembly is well-suited for interference detection, and possibly very useful for future reference. However, with the initial goal of creating a simulation for this assignment, certain simplifications have been made to the model of the chassis. Those simplifications will (almost certainly) not affect the behavior of the final model for this specific test, for example, a plate positioned in the longitudinal rear beam to reinforce the torsion bar mounting point has been excluded from the design because it is outside the area affected by the applied force (see Figure S21).

Overall, the chassis is easily replicated in SolidWorks due its simplicity and straight edges. However, the front section, where the engine and gearbox are positioned, is more complicated due to various angles and curves and cut-outs. To aid the designing process for this section of the model, replacement members for chassis restauration were provided by R-Quatre.nl allowing for precise measurements (see Figure S22). Additionally, inspecting an uncoated chassis revealed the location and quantity of spotwelds used to attach all parts together (see Figure S23 and Figure S24).

Load case

Similar to the situation depicted in Figure 3, the load case chosen for the simulation in this project is based on a vehicle driving over a large bump in the road. When driving over a speed bump with considerable speed, a so-called bump-factor of 3G, three times the gravitational constant, can be applied to the weight acting on the structure. [4]

Keeping in mind that a real-life validation test had to be performed, a heavy weight was available in the HAN storeroom: a steel block of 581 kg (Figures S43). Conveniently, a driver and passenger of approximately 80 kg and two car seats of each 15 kg amount almost exactly to one third (1G) of the available weight. Furthermore, to equally spread the weight over the central lateral beam (and simultaneously prevent indentation damage to the chassis), a wooden beam covering exactly the dimensions of the central beam was used to distribute the force (see Figure 16 and Figures S43). The wooden beam weights 5.8 kg:

Total applied weight:

581 + 5.8 = 586.8 kg

Total force:

586.8 * 9.81 = 5756.5 N

Weight driver, passenger & seats:

***586.8 / 3 = 195.6 kg**:* *(80 + 15) * 2 = 190 ≈ 195.6 kg*

Although, in reality the load of such a bump impact is not solely absorbed by the chassis of the vehicle, but also the vehicle body. As can be derived from the formulas from the second polar moment of area, stiffness (resistance to bending) of an object is increased by increasing the distance from the central axis to its edge; in the case of the Renault 4, the roof structure made of sheet metal contributes significantly to the overall stiffness of the vehicle (see Figure 5).



Figure 5: A longitudinal cut-through of the Renault 4. [5]

The Renault Rodeo

To simplify this simulation and prevent the modeling of the entire vehicle body, the assumption could be made that in this case the vehicle of interest is a Renault Rodeo rather than a Renault 4, considering that it was built on the exact same chassis (see Figure 6). The Renault Rodeo does not have a solid roof, and the thinnest section of its fiber glass reinforced plastic tub is exactly located at the central lateral beam which therefore does not contribute to stiffness much. Similarly to the Renault 4, both seats in the rodeo (no rear couch present) are mounted on the central lateral beam; as per example in Figure 5. The previously calculated load was therefore applied to the central lateral beam.



Figure 6: A 1975 Renault Rodeo 6, built on the Renault 4 chassis.

Simulation

This section of the report discusses the approach and settings of the simulation of the simulation in SolidWorks. After concluding the geometry of the chassis model, the next step was establishing a correct simulation set-up, this includes the definition of part types, connections, fixtures, forces, and the mesh.

Set-up

Since the platform chassis of the Renault 4 is made of sheet metal, the choice has been made to simulate the individual chassis parts with surfaces, which then are converted to shell elements by only utilizing the inside or outside face of the part. Utilizing shell elements, which are infinitely thin, reduces calculation time with respect to simulating solid parts that incorporate the thickness of the sheet. However, the simulation still requires an indication of the thickness and offset dimensions of each part. Thickness offset is crucial, this can best be explained with Figure 7: When the thickness offset is rendered in the wrong direction, the outer dimensions change, resulting in a different (larger or smaller) cross section of the beam, thus developing a stiffer or weaker part than expected. Furthermore, in an assembly this can lead to interfering parts. The shell element and offset settings can be seen in Figure S25, Figure S26, and Figure S34.

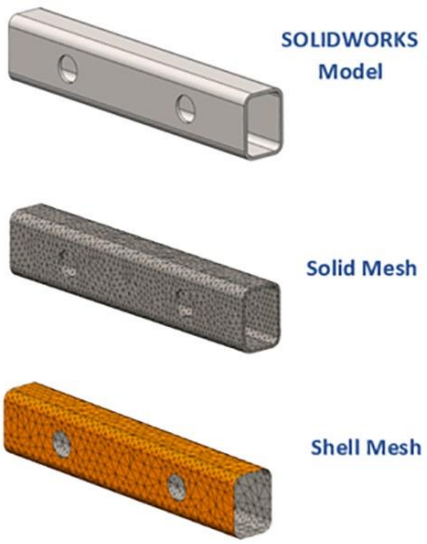


Figure 7: SolidWorks part types. [10]

To stay as close as possible to reality, all individual parts in the assembly of the chassis model have been attached to each other using the spotwelding feature in SolidWorks. For this, an uncoated chassis and replacement chassis members have been inspected, revealing the quantity and location of the spotweld in the Renault 4 chassis. In Figure S27 the application of the spotwelds for the simulation can be seen. Additionally, experimentation was done with the connection of the suspension brackets (front and back) to the chassis members by using bolts or pins, this will be explained in detail in the next sub-chapter.

Fixtures are used to restrict movement of parts or part features in specific directions or rotations. Immovable fixtures in the X, Y, and Z-direction can for example be applied to the touching points of the suspension brackets (see Figure S31), similar to reality (see Figure S45 and Figure S46). Additionally, the next sub-chapter contains an explanation on a more advanced fixture method based on geometry symmetry.

Arguably one of the most important features of the simulation set-up is the introduction of the force acting on the chassis. As can be seen in Figure S29, a force is applied to the edges of the central lateral beam, resulting in a situation that can be replicated in the real-life test.

The last step in preparation for the simulation is dividing the model into many small sections at which interaction points, nodes, the software performs multiple computations. The size of these sections, so called mesh size, influences the accuracy of a simulation. With a finer mesh size (more & smaller sections) results in a more accurate outcome, however it increases computation time, and vice versa for a coarser mesh. See Figure S34.

Iterative updates

As to be expected, the first few iterations of the simulation still contained small errors. In this section of the report the reasoning behind the iterative updates throughout the simulations is explained.

In the first attempt of the simulation, the surface model of the chassis, already prepared as shell elements, was combined with solid suspension brackets which were determined to be rigid, aiming to reduce calculation time by not calculating the deformation of the thicker (2-3.5 mm) suspension brackets opposed to the thinner chassis (0.8 mm). Unfortunately, this simulation failed due to a memory constraint of the computer; the number of accounted nodes was too high.

The next attempt was based on the reduction of total number of nodes. Although the chassis is in fact not symmetric due to the double torsion bar rear axle design, by halving the model, substantially fewer nodes could be attained. Unfortunately, the effect on the total deflection by the slight wheelbase difference had to be disregarded. A symmetrical boundary condition was applied to the model, indicating that the interactions on the model (nodes) on the cutting plane are mirrored from left to right. Important to note is that the applied force now should be halved as well (2878.254 N). An additional measure to reduce the number of nodes, thus the size of the required memory, was the elimination of the left side of the front beam, and the rear part of the trunk-supporting longitudinal beam as those sections do not contribute to the stiffness in this situation. The remaining part of the chassis for the simulation can be seen in Figure 8 below.

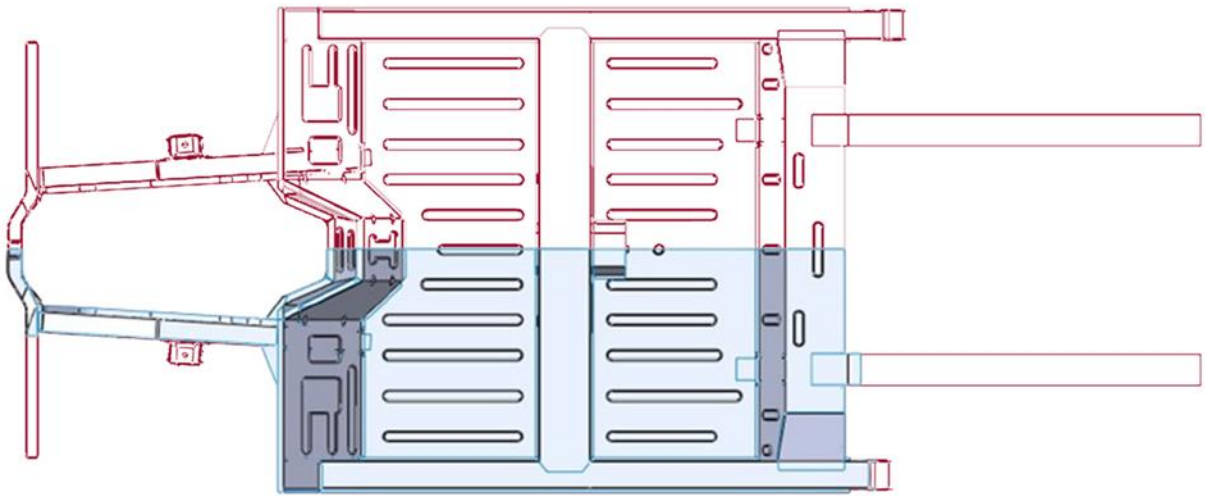


Figure 8: Symmetrical reduction of the chassis model for simulation.

Although the aforementioned measures, plus the elimination of the suspension support brackets successfully reduced the required memory size, the simulation still deviated from reality. The symmetric boundary condition constrains any displacement normal to the reference plane. The part, or rather the cut line of the symmetrical part, can slide over the reference plane but not penetrate or move away from it. However, the symmetry feature of SolidWorks does not consider first displacement derivatives or discontinuous rotations along the reference plane. This therefore provided inaccurate simulation results and required further attention. [6]

In the previous iteration of the simulation the suspension brackets, attached with bolted connections, were excluded from the simulation because of an error with the bolt pre-load, and subsequent issues with bolt contact and relative length after updating the suspension brackets from solid to surface models. The issue with the symmetrical boundary was therefore found with a model in which the fixtures were not representative of reality: translational fixation on the edges of the bolt holes of the suspension brackets, in essence fixating the entire front section and rear axle attachment point.

In the following iteration of the simulation, the symmetrical boundary condition was replaced by a so-called 'advanced fixture'. This method included the limitation of rotations along the reference plane, as well as the previous restriction of nodes through the mirror plane similarly to the symmetric boundary condition (see Figure S30). This provided confirmation of the correct working of the advanced fixture method.

In the final simulation iteration, the suspension brackets were re-introduced as surface models (shell elements), but now using pin connectors rather than bolted connections (see Figure S28). Although this deviates from reality, in theory it should yield the exact same result for this specific load case: since there is only a vertical force introduced, there can never be an axial reaction force provoked on the horizontally positioned pins.

Judging by the simulation animation provided by SolidWorks, the final iteration of the simulation was an improvement with respect to the last attempt. In Figure 9 below, the deformation of the chassis is presented using colors. Where zero deflection is blue, and the maximum deflection, of about 5.07 mm, is red. At the point in the middle of the central lateral member, the simulated deflection was 4.070 mm.

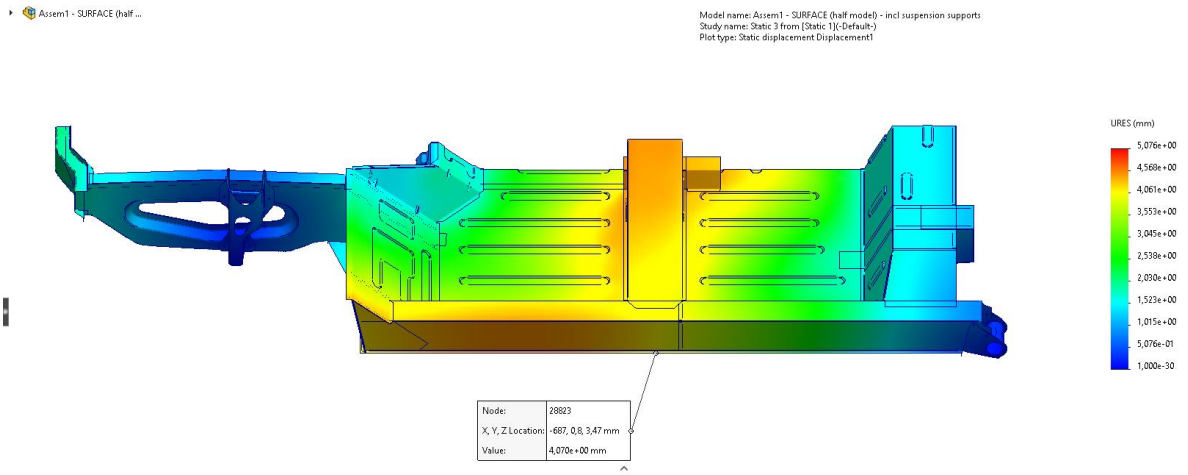


Figure 9: The simulation provides a deflection of 4.070 mm at the same point of measurement as in the validation tests.

However, the final iteration of the study further advances to a realistic representation are made by removing the Z direction fixture of the rear supports. By including fixations in the Z direction for edges on both the front and rear supports, the total stiffness of the chassis is increased due to tension in the chassis. The tension of the total chassis affects the measurement since the supports cannot slide in the Z direction. Allowing one of the supports to slide in the Z direction represents reality better. See Figure S32 and Figure S33.

Results

In the final iteration of the simulation, the total deflection measured on the side of the central lateral beam amounts to 4.725 mm; approximately 4.73 mm. In the appendix, Figure S35 to Figure S42 provide more information on the deformation (visible & scaled) from each side of the chassis. Stiffness of the chassis can be calculated by dividing the force by the deflection [7];

5756.5 N / 4.725 mm = 1218.3 N/mm

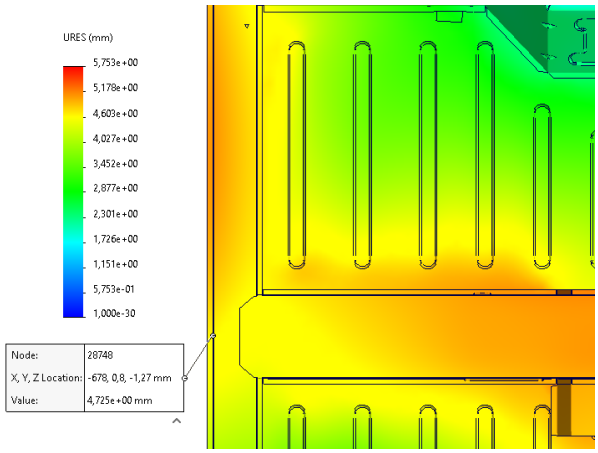


Figure 10: Measured deflection of last iteration.

Singularity check

A stress singularity is an unrealistic stress concentration, exceeding stress values in the surrounding regions, due to inaccuracies within the FEA calculations. The key characteristic of a singularity is the rise to infinite stress with an increase in mesh size. However, stress singularities occur at locations where natural stress concentrations will occur, such as sharp corners and bolt holes, see Figure 11. Therefore, the so-called stress hotspots need to be checked to determine if it is a stress singularity. This can be done by verifying whether the stress rises without convergence (reaching infinity) by increasing the mesh density.

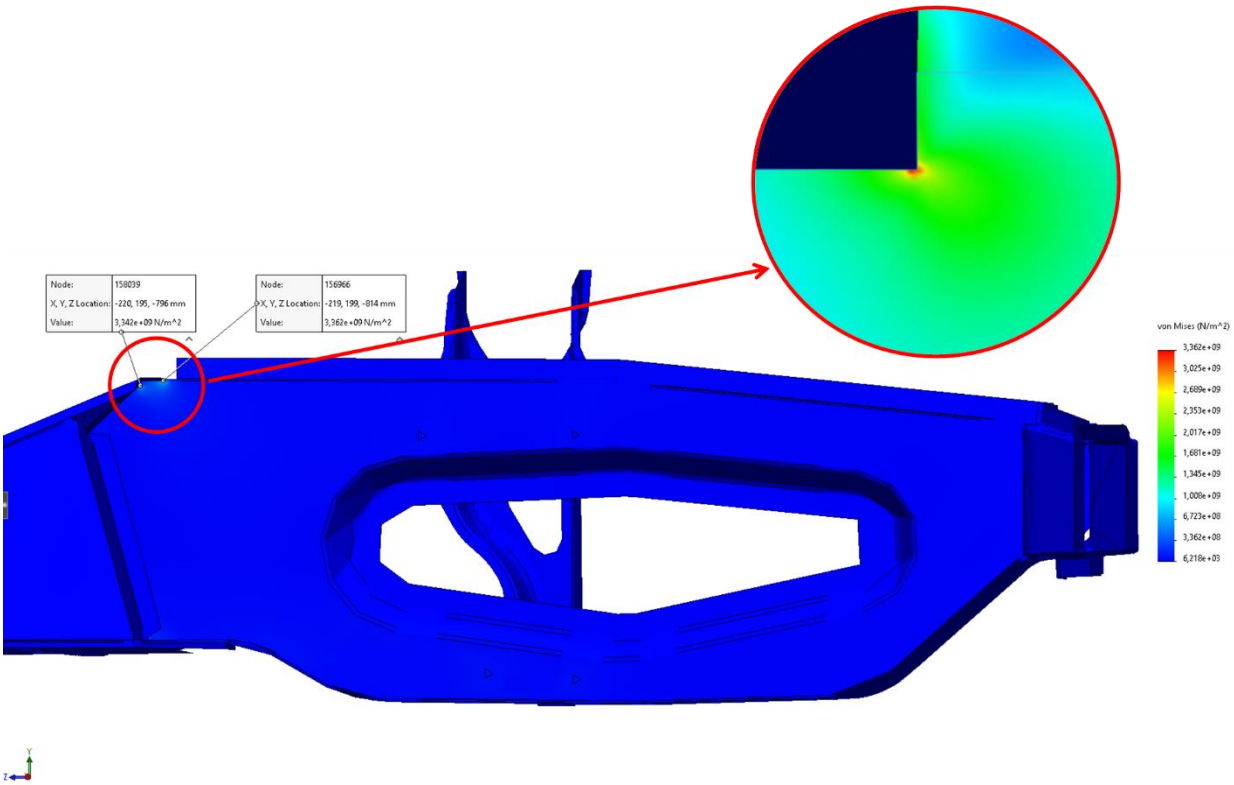


Figure 11: Stress concentrations in two locations of the front section of the chassis.

In the graph (see Figure 12), two stress points are checked for singularities. The graphs, depicting two invested stress concentrations, appear to approach infinity and therefore the assumption can be made that stress singularities are present. The attempt was to simulate with a local mesh (using *split lines*, Figure 13 and Figure 14) of 0.004 mm, but as every step requires increased computing power, this final step failed due to insufficient memory (simulating for 6+ hours before failure). However, the stress seems to be increasing non-linearly suggesting its approach to infinity with a decrease in mesh size.

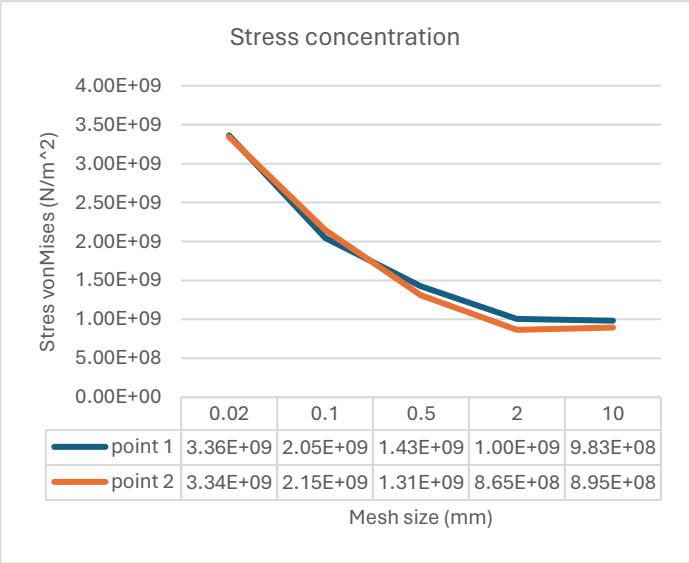


Figure 12: Stress singularity investigation.

As mentioned before, a stress singularity is usually found near sharp corners or bolt holes. A stress singularity should not be ignored as it could well be an actual design flaw. In reality, the chassis does not contain infinitely sharp corners in its sheet metal.

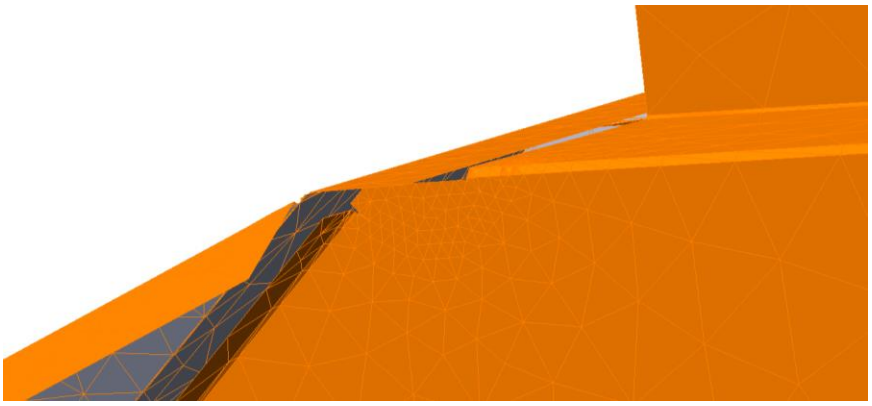


Figure 13: Mesh size at stress concentration (front section) reduced to 2 mm.

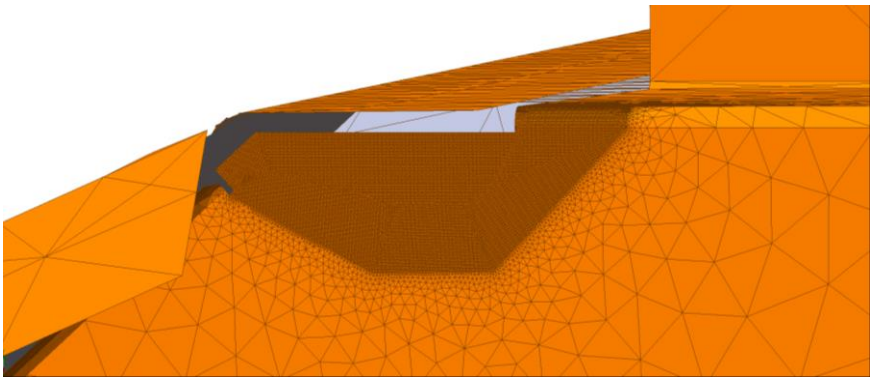


Figure 14: Mesh size at stress concentration (front section) reduced to 0.1 mm. The reduction of mesh size makes the split line section clearly visible.

Validation test

Validating the credibility of the simulation can be done by replicating with a real-life test and comparing the results to the simulation results. It is important to precisely mimic the simulation with the real-life test to prevent circumstantial differences that possibly affect the results. This includes, most significantly, an equal amount of applied weight and support points that accurately represent the fixtures of the simulation.

Set-up

Conveniently, as previously introduced, a weight of exactly 581 kg was available (*see* Figures S43). A large block consisting of 10 steel columns welded together, representing roughly the force applied on the central beam considering a combined weight of two passengers and seats on the central lateral beam when subjected to a 3G constant. Since weights in this mass range are not readily available, the steel block present at the HAN storage was the deciding factor for the applied force in the simulation, while simultaneously corresponding quite well to a realistic situation. Furthermore, the block was already equipped with two towing eyes: facilitating the transportation and handling. As can be seen in the picture below, the block was attached to the overhead crane (a.k.a. bridge crane) in the ARLA where the validation test was performed.

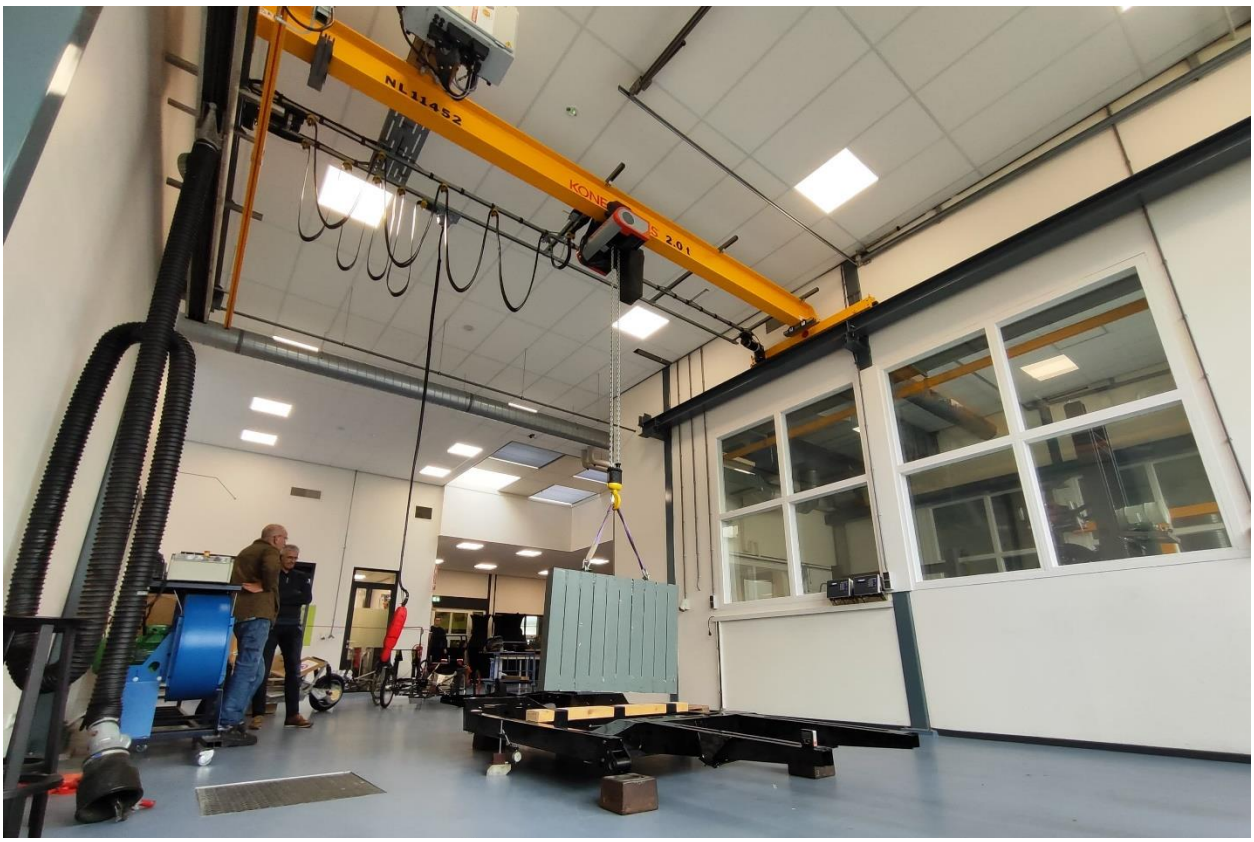


Figure 15: The validation test set-up including the overhead crane in ARLA used for lifting the weight.

As in the simulation was chosen for a distributed load, a beam was utilized to spread the load of the compact steel block over the full surface area of the central lateral chassis beam. This simultaneously acted as a protection of the brand-new chassis against scratches and dents. The weight of the wooden beam was approximately 5.8 kg, making the total sum of load deflecting the chassis approximately 586.8 kilogram (5756.5 Newton).

Having considered the support options for the validation test before the simulation, the choice was made to include the original suspension system support brackets of the Renault 4. The contact of the rear axle supports is simply a single lateral edge, allowing for rotation in the XY plane (Z-direction = longitudinal), see Figure S46 of the rear supports. Even though the aforementioned rotation in the XY plane is the expected outcome of this 3-point bend test and ideally supports should be considered allowing for this rotation, due to the geometry of the front suspension support brackets and the aim for accurate comparability between the simulation and validation test, chosen was for simple supports at a well-defined position that however do not allow for rotation in the XY plane, but do allow for rotation in YZ. See Figure S45 of the front supports in the appendix.

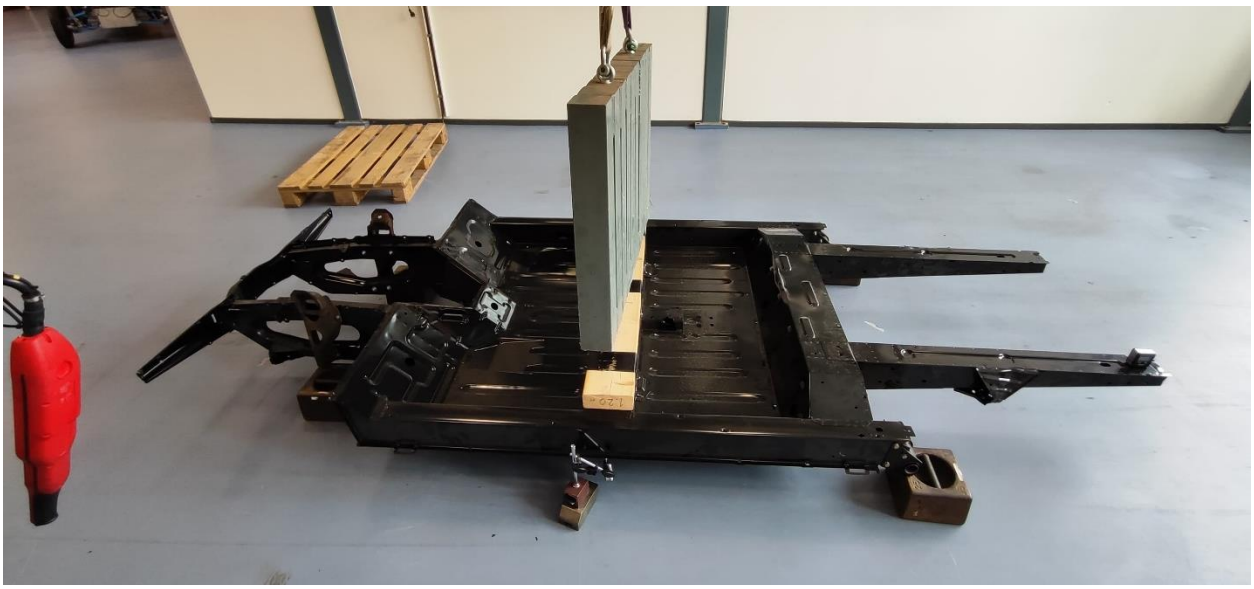


Figure 16: Validation test set-up, chassis positioned on four solid steel blocks, and a dial gauge.

Furthermore, contrary to the simulation where no real 'floor' surface is present, the total installation height of the validation test set-up is increased with solid steel *blocks* (Figure 16). Allowing for the deflection of the chassis, the increased height of the set-up also facilitates the placement and measuring of the dial gauges. The positioning of the dial gauges reading the maximum deflection of the validation test is also trivial. A point in the middle of the central lateral beam is defined *as* 790 mm measured from the front edge of the main part of the chassis. On both sides of the chassis the deflection is measured with dial gauges.

With a slight pre-tension the dial gauges are positioned underneath the chassis on both sides. Opposed to the simulation, this is a non-perfect situation where slight changes are bound to happen, such as the settling of the set-up after applying the weight for the first time. Therefore, rather than measuring the positive deflection, the retraction of the elastic deformation is utilized to determine the average deflection of the chassis.

Results

In the table below the readings of both dial gauges have been sorted in a table, as previously mentioned, the pre-tension on the dial gauge needles dictates in this case that the measurements start at a reading of 4.00 mm. After a maximum deflection of 5.20 mm (left), lifting the weight did not result in a total retraction of the same amount. This difference is likely the settling of the position of the chassis on the four steel support blocks.

Measurement		**Deflection**				
left	right		left		right	
4.00	4.00		deflection	retraction	deflection	retraction
9.20	10.36		5.20	4.84	6.36	5.01
4.36	5.35					
9.56	10.18		5.20	4.78	4.83	4.94
4.78	5.24					
9.52	10.04		4.74	4.68	4.80	4.83
4.84	5.21					
9.63	10.06		4.79	4.73	4.85	4.90
4.90	5.16					
			Average			
			left		right	
			4.98	**4.76**	**5.21**	**4.92**

Table 1: Validation test deflection measurements.

To conclude, on average the deflection is 4.76 mm and 4.92 mm on the left and right respectively due to applying a weight comparable to a bump impact situation considering a constant of 3G. The lateral difference in deflection of 0.16 mm is likely due to the difference in wheelbase; 48 mm longer on the right side inherent to the double torsion bar suspension system of the Renault 4. Unfortunately, this observation could not be reflected in the simulation results due to the size constraint requiring a symmetric boundary condition and thus eliminating the small difference in wheelbase length.

Discussion

Comparing the simulation to the real-life validation test, the results are quite close: 4.73 mm to 4.76 mm respectively, a difference of only 0.03 mm. It represents 0.63% of the measured deflection. This error could be caused by various factors, such as chosen material, the exact location and quantity of spot welds, but also the measurement equipment and reading error. Overall, the simulation results are considerably close to reality, and the model will therefore be useful for future simulations; for example, locating stress concentration points in the chassis under deformation, or investigating the stiffness increase of the assembly by introducing a safety cage.

Safety cage



Figure 17: Roll cage for a Renault 4 (designed to FIA specifications) installation with friends.

Having proven the accuracy of the model and the simulation, a next step can be taken. With the aim of building a rally car and learning as much as possible during the endeavor, performing a simulation to find out the increased stiffness of the added safety cage fits this project perfectly. A model of the safety cage was already available from previous design efforts to determine the amount of material required to build the structure and the additional weight to the vehicle. This made the simulation process easier; the design was converted to beam-type elements to reduce calculation time. *Beams,* other than *solids,* represent structures with a single line, over which its cut-through profile is extruded. This furthermore helps with automation and ease of modelling.

Since the chassis model for the simulation consists solely of shell elements, the connection of the safety cage (beam elements) to the chassis needs to be manually adjusted as automatic contact sets usually do not work when combining beam and shell elements. In this case, local interactions sets are generated between the end point of the safety cage tubes and their foot plates, and additional local interactions sets for the foot plates to the chassis. In Figure 18 and Figure 19, the deflection of the total assembly can be seen, for this simulation the longitudinal beams towards the rear had to be extended slightly, and the missing spotwelds had to be introduced again. With the addition of the safety cage, the simulation indicates a total deflection of 3.77 mm measured at the central lateral beam. The increased stiffness can therefore be calculated:

5756.5 / 3.767 mm = 1528.1 N/mm

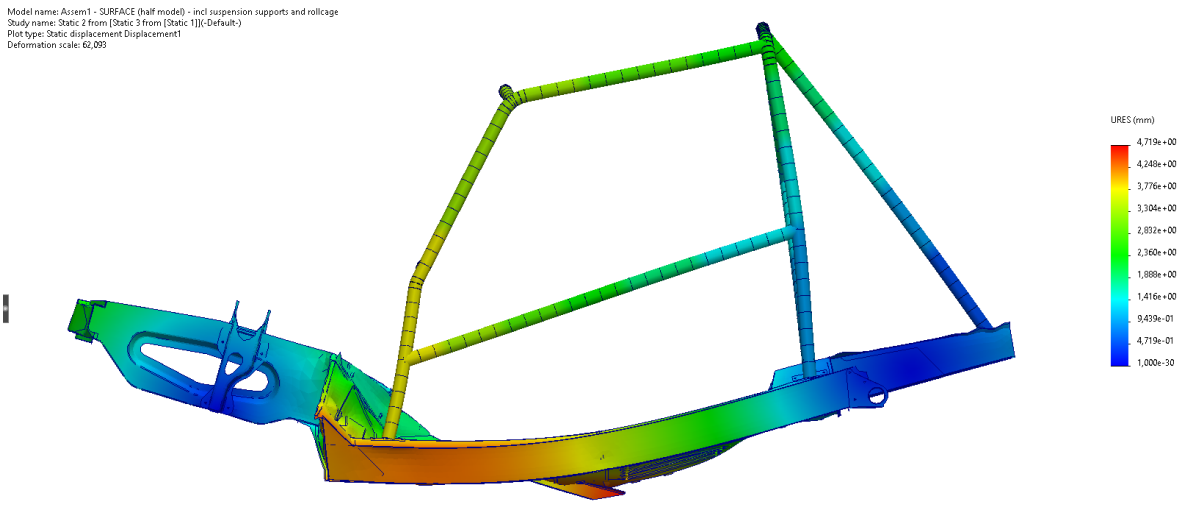


Figure 18: Simulation of stiffness increase with safety cage installed.

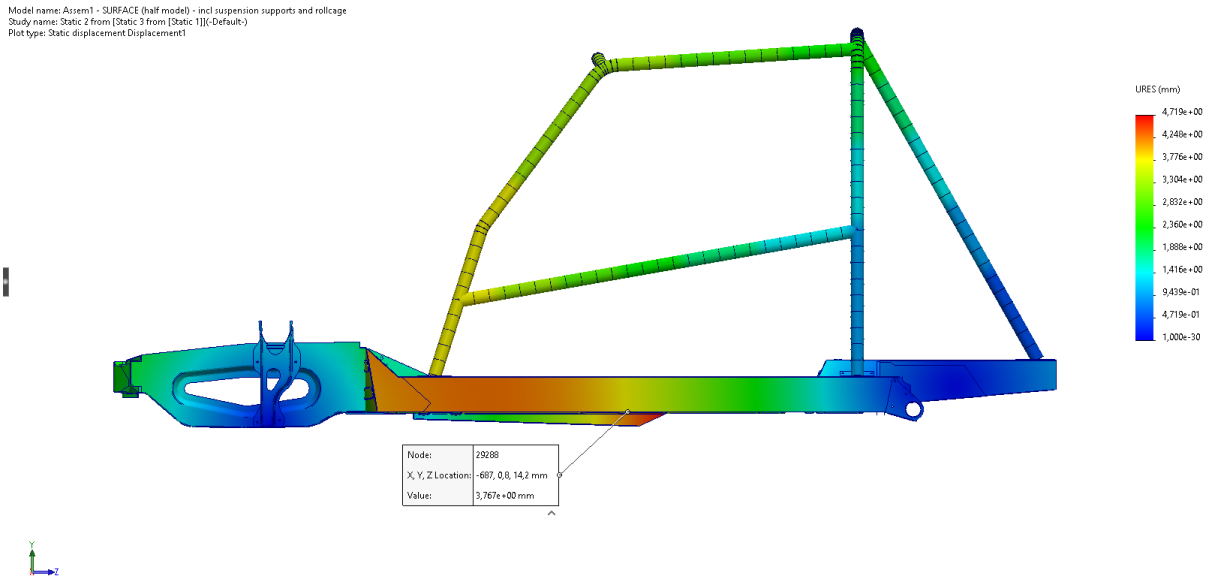


Figure 19: Total deflection of the assembly measured at central lateral beam.

Conclusion

After constructing the SolidWorks model of the chassis in as much resemblance of reality as possible, the deviation between the final iteration of the ongoing improvements of the simulation presented a deflection value within 0.63% of the validation test; a total simulated deflection of 4.73 mm versus 4.76 mm measured deflection.

Additionally, the investigation of stress concentrations by increasing mesh density and measuring the maximum stress brought forward probable stress singularities.

And finally, after assuming correctness of the model and simulation, a safety cage was introduced to figure out the increase in stiffness of the assembly. Although the decrease in deflection was expected to be higher, the stiffness increase with the introduction of the safety cage was 125.4% of the original stiffness without safety cage.

Acknowledgement

As previously mentioned, this project was sponsored by R-Quatre.nl providing a brand-new chassis and its transportation to the HAN in Arnhem for the validation test. The expertise of Jan-Willem van Nifterick was fundamental to the creation of the Renault 4 chassis model in SolidWorks. Furthermore, aiding with providing photographs, measurements, replacement chassis members for detailed inspection to accurately model the chassis.

Additionally, a special thanks goes out to Appie Peters and Wolbert van Dronkelaar for their assistance in preparing the test set-up and handling the weight during the validation test.



Figure 20: Wolbert and Appie helping with the validation test.

Appendix



Figure S21: structural reinforcement inside the longitudinal rear beams. [8]



Figure S22: Replacement chassis members provided by R-Quatre.nl used for reference measurements and determination of the position and number of spot welds.



Figure S23: Picture of the underside of an uncoated chassis with visible spot welds. [9]

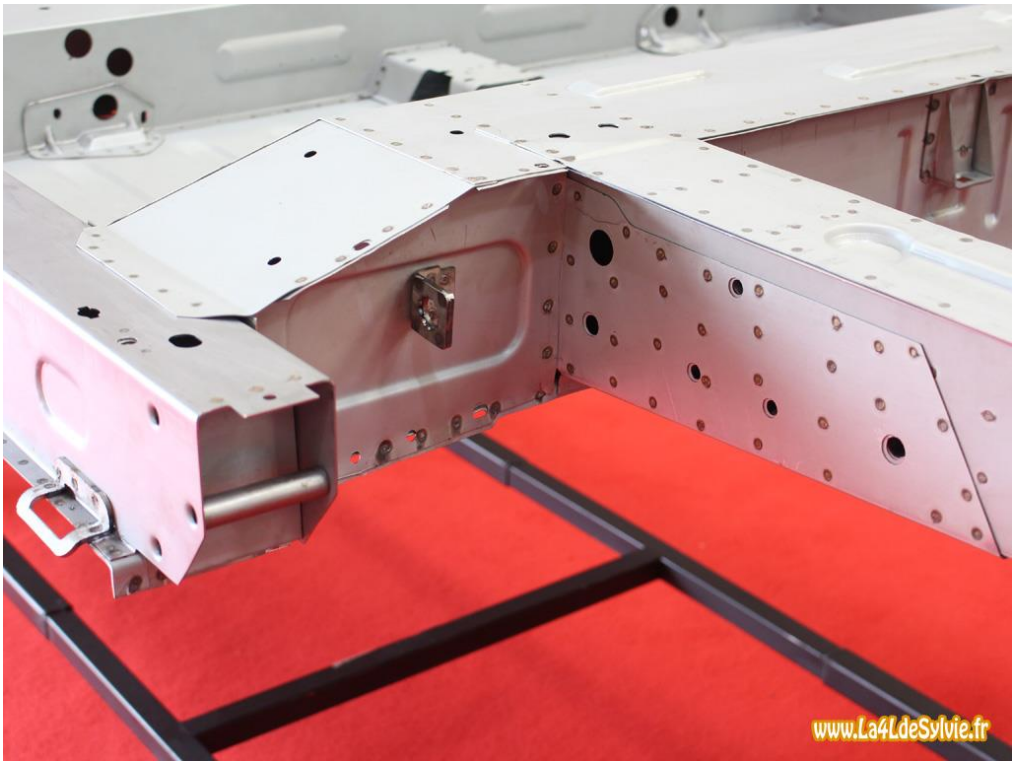


Figure S24: detailed picture of the chassis' rear section and the visible spot welds. [9]

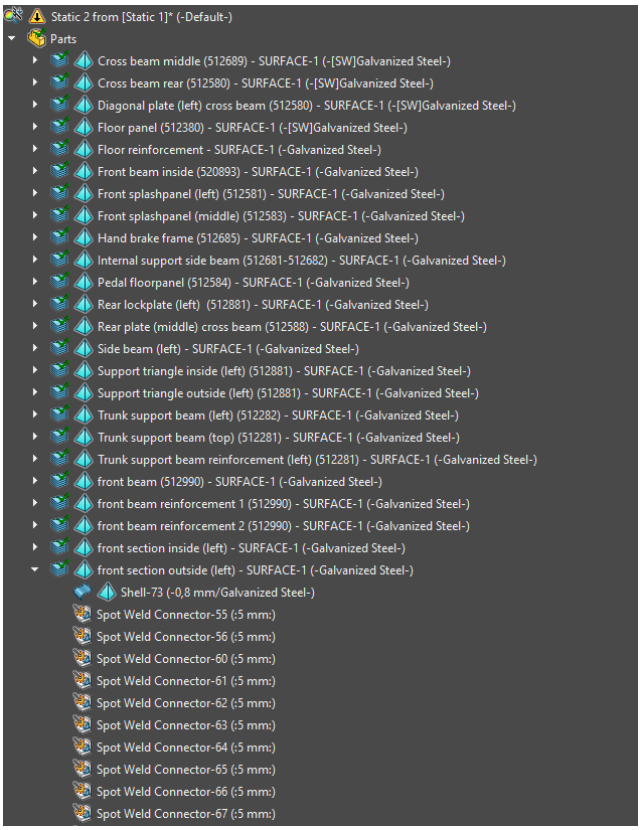


Figure S25: Full list of chassis parts prepared and approved for the half-model simulation. The final part (front section outside (left)) is unfolded; revealing shell elements, applied thickness and material.

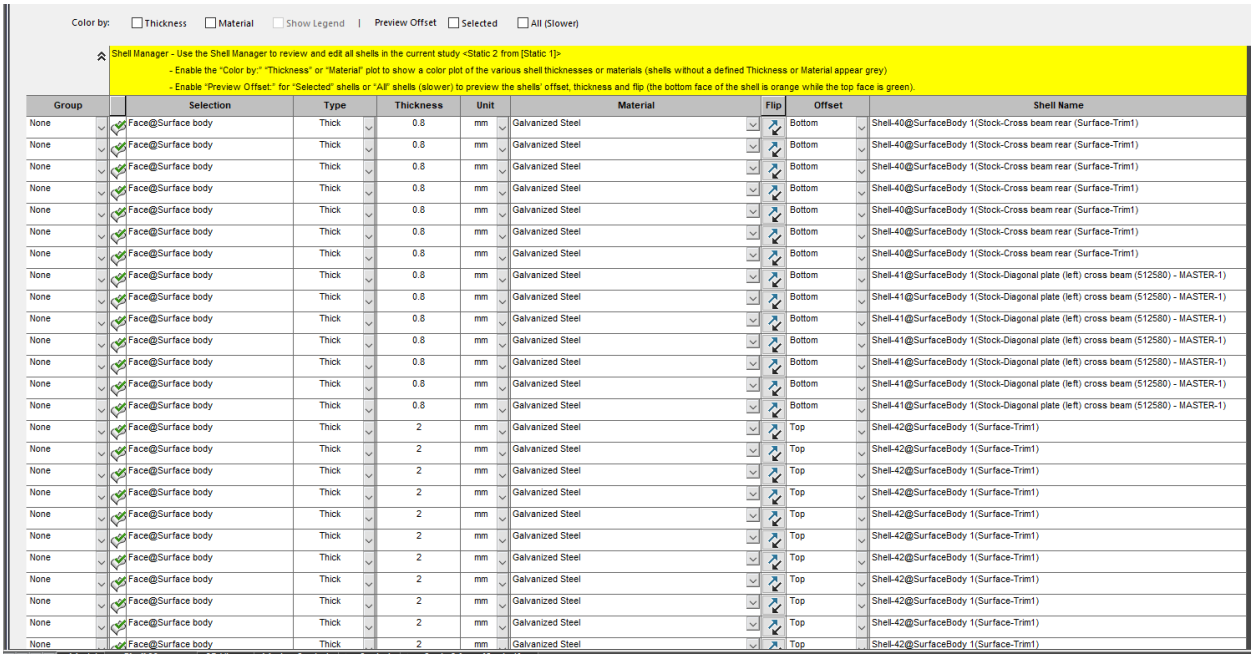


Figure S26: An overview provided by the shell manager, indicating the shell type, thickness, material offset, and name of each surface body.

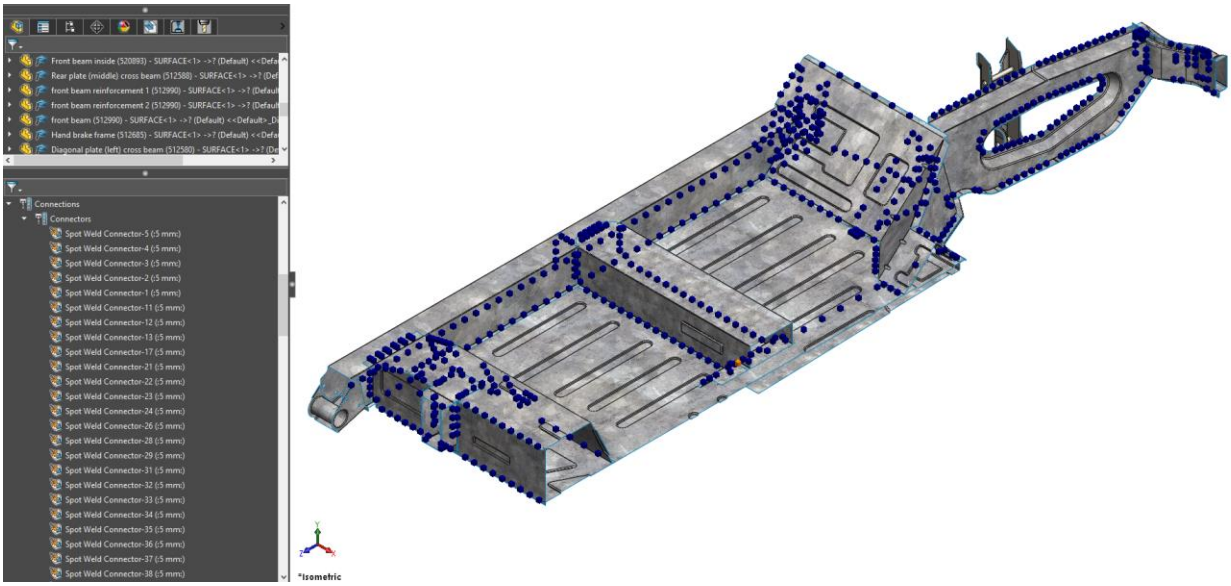


Figure S27: Visualization of spot weld locations represented by the blue dots.

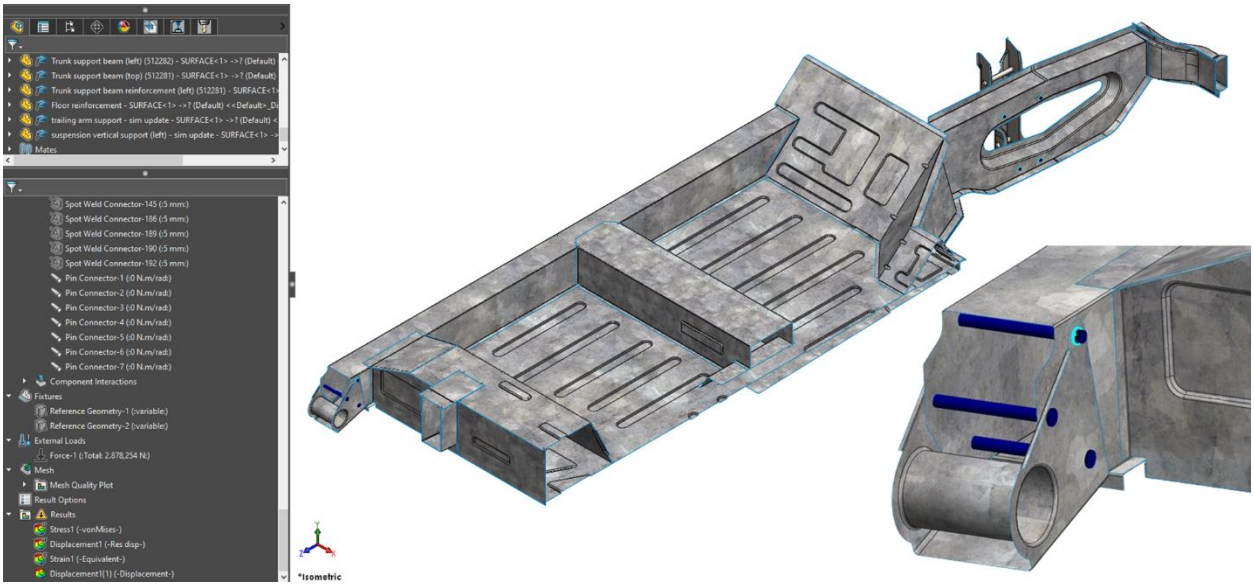


Figure S28: Pin connections between the suspension brackets and the chassis assembly.

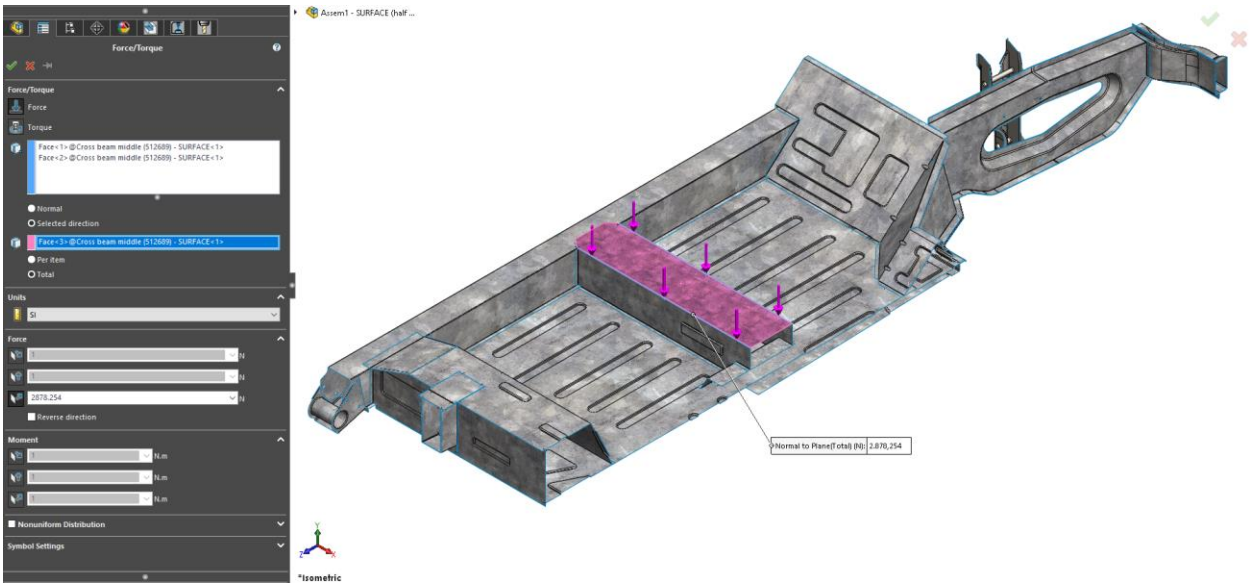


Figure S29: Purple arrows indicating the application of the force (2878.254 N: 293.5 kg) on the two edges of the vertical surfaces with respective direction according to a reference plane.

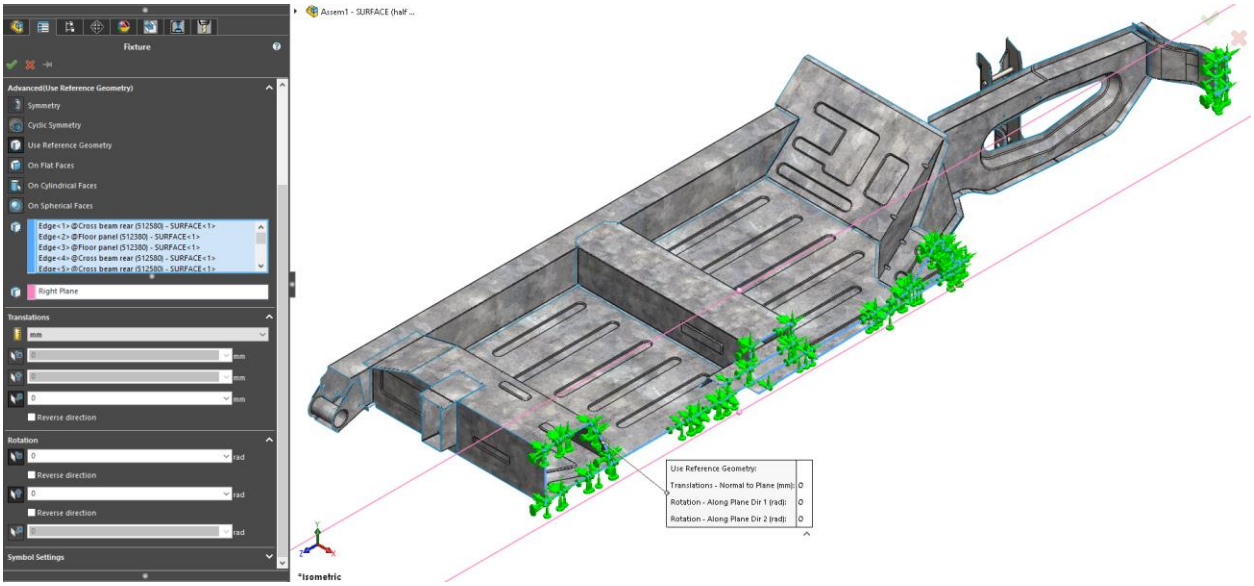


Figure S30: The set-up of the correct symmetry simplification using an advanced fixture feature: reference geometry (mirror plane).

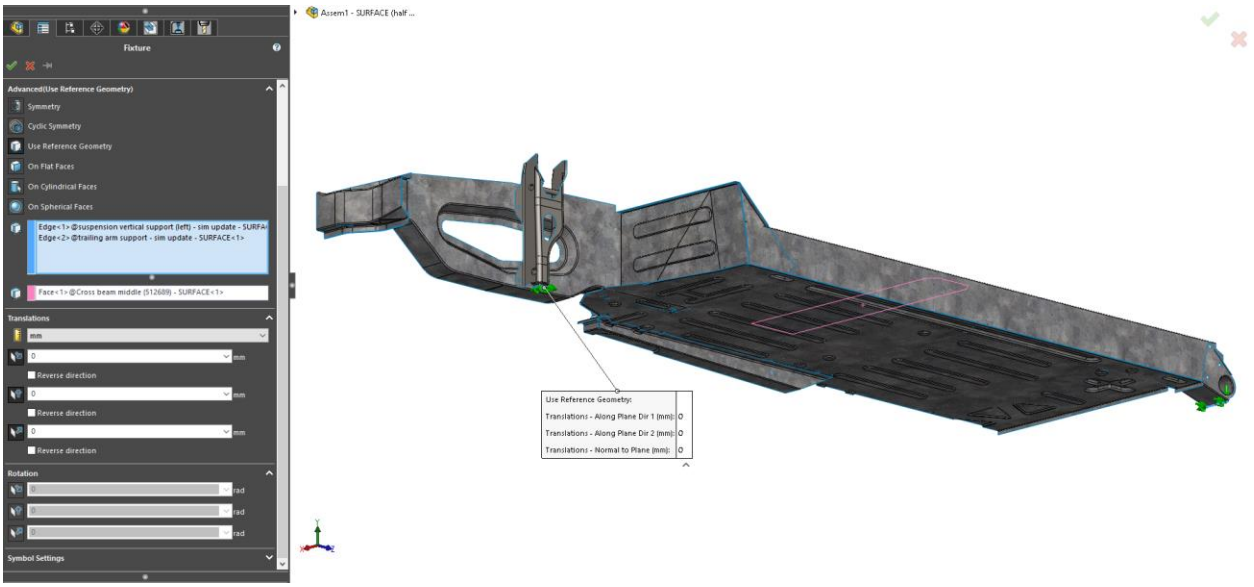


Figure S31: Green arrows indicating the fixtures at the support bracket bolt holes. (V1)

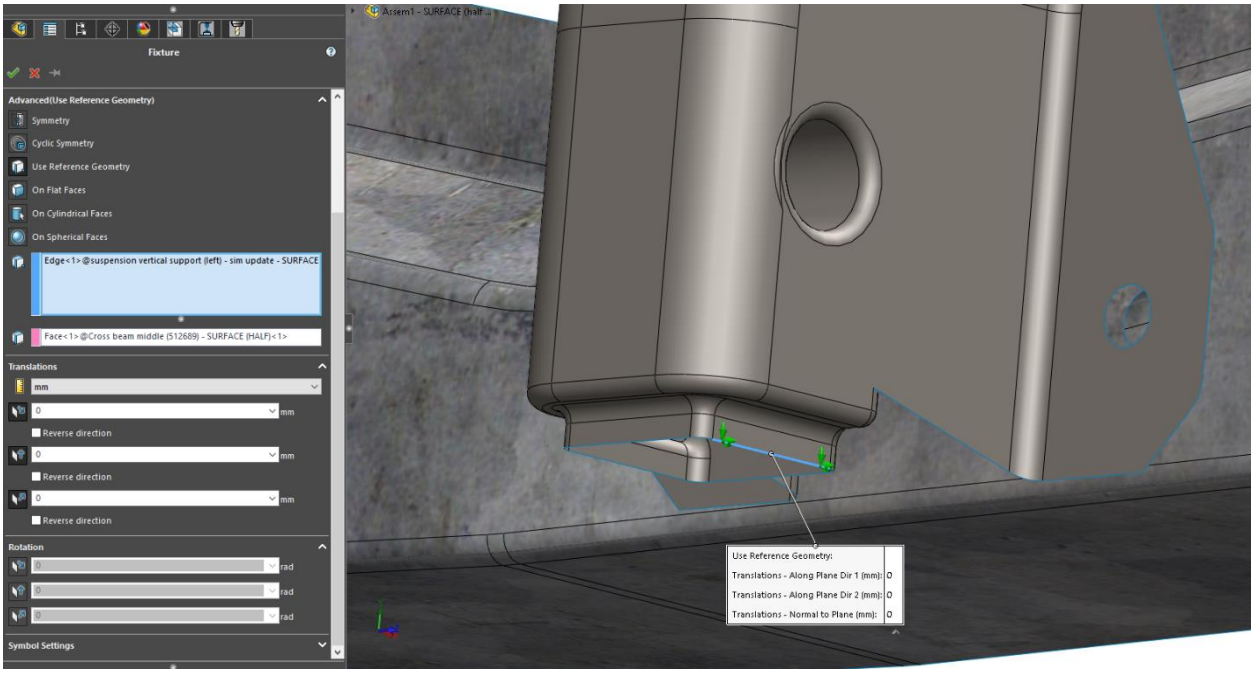


Figure S32: Close-up of front support fixture (edge type). Fixed in X-, Y-, and Z direction.

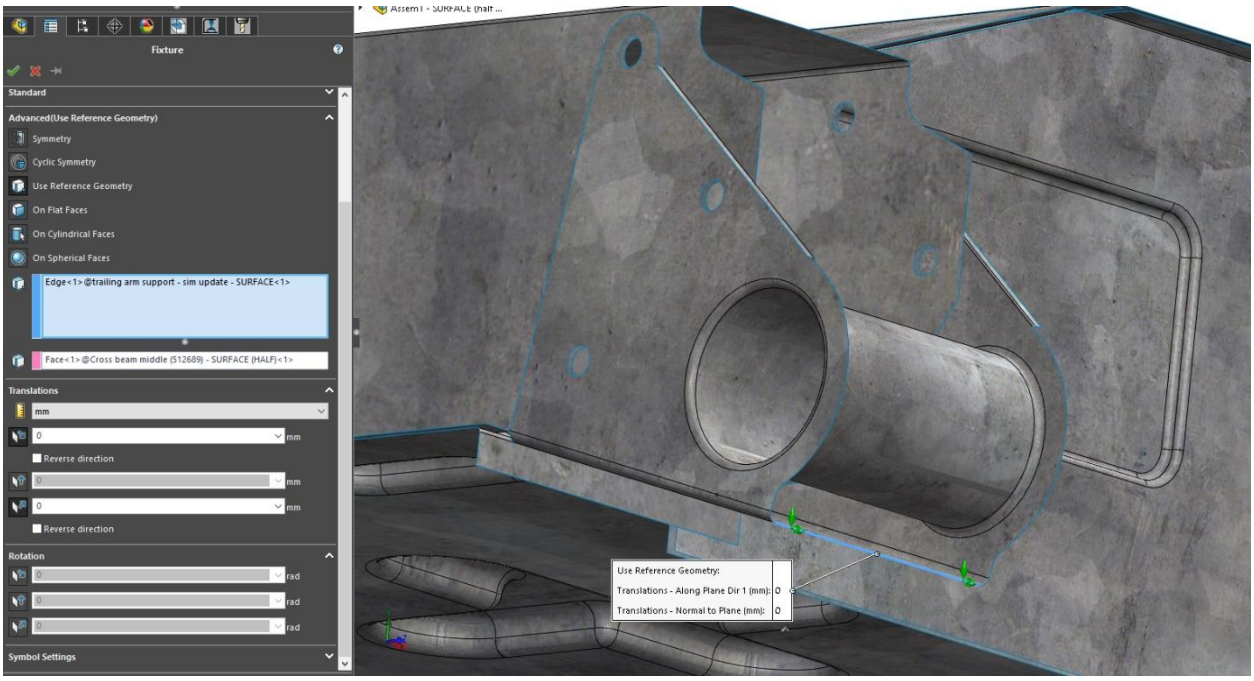


Figure S33: Updated fixture (X- and Y direction) for rear support (free in Z direction: V2)

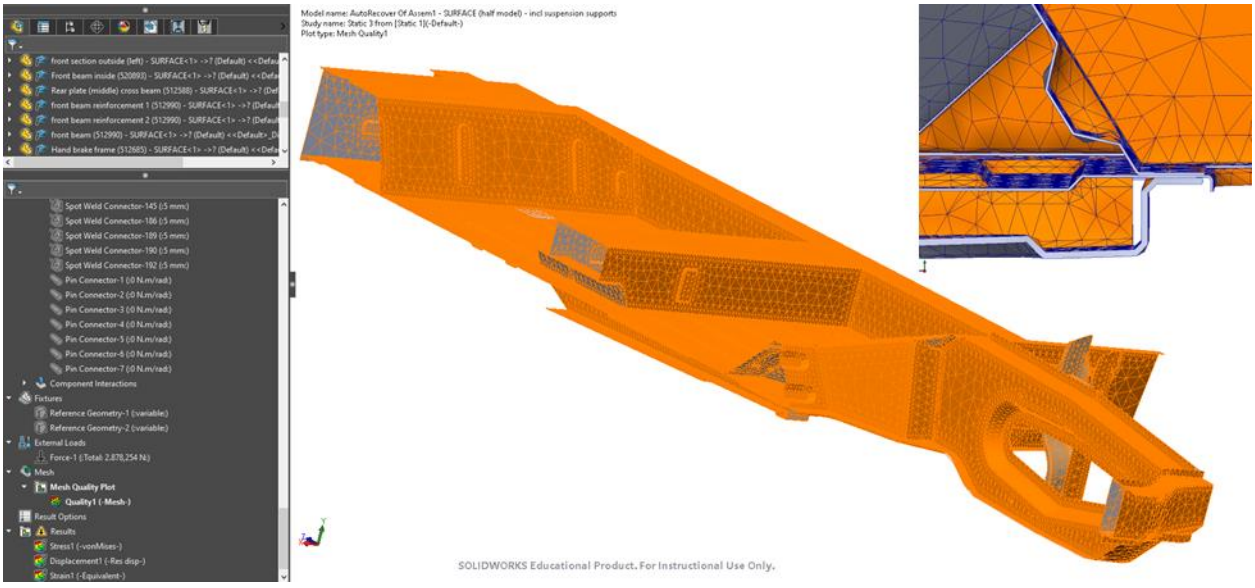


Figure S34: Mesh settings including, density and other parameters. Additionally, the simulated thickness offset direction is presented by the orange and grey faces.

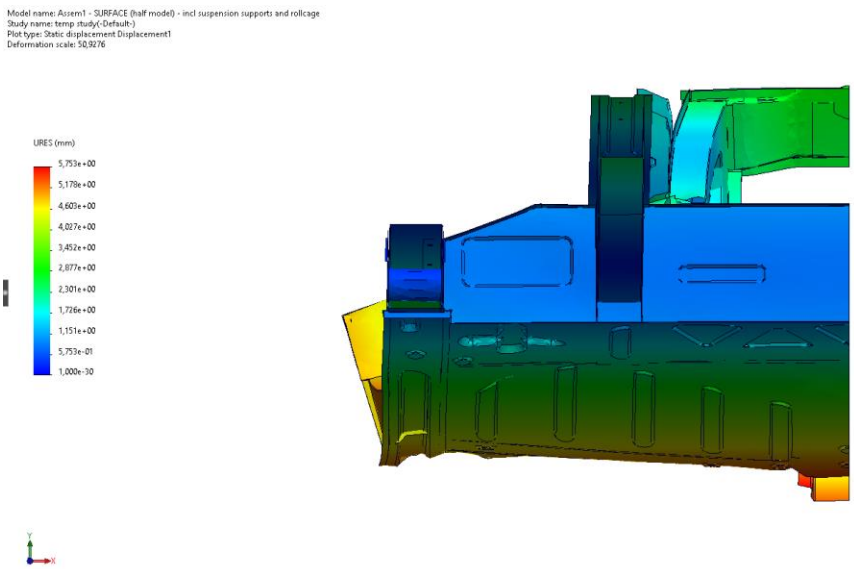


Figure S35: Rear view of deformed chassis. (scaling factor ±50x)

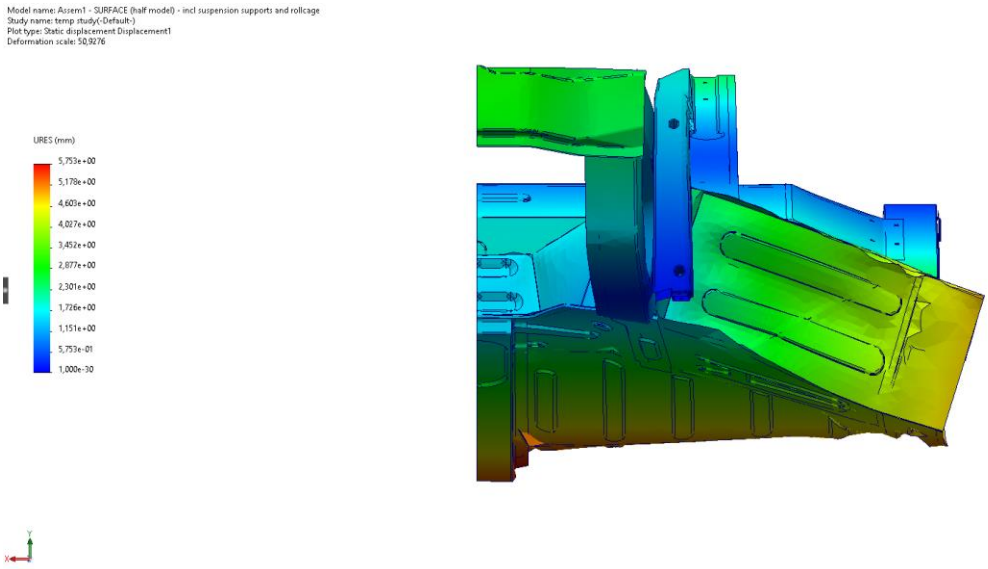


Figure S36: Front view of deformed chassis. (scaling factor ±50x)

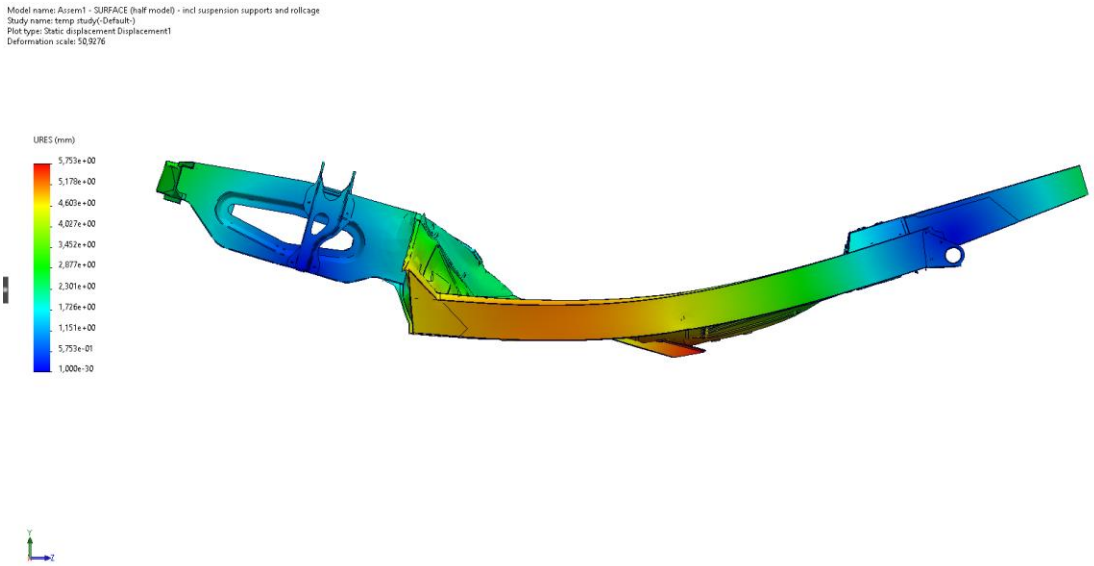


Figure S37: Left side view of deformed chassis. (scaling factor ±50x)

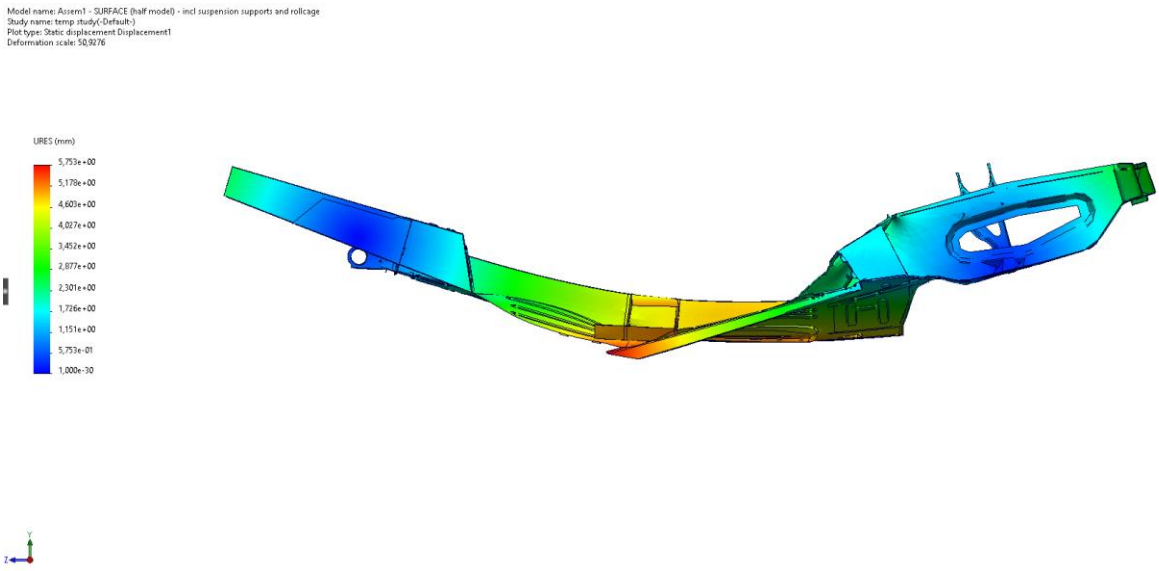


Figure S38: Right side view of deformed chassis. (scaling factor ±50x)

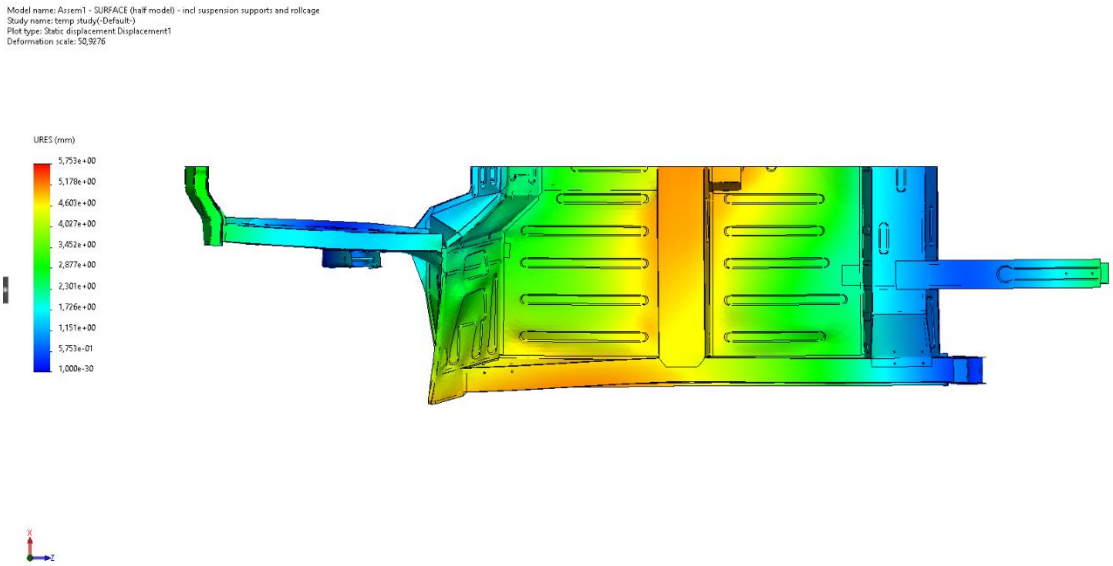


Figure S39: Top view of deformed chassis. (scaling factor ±50x)

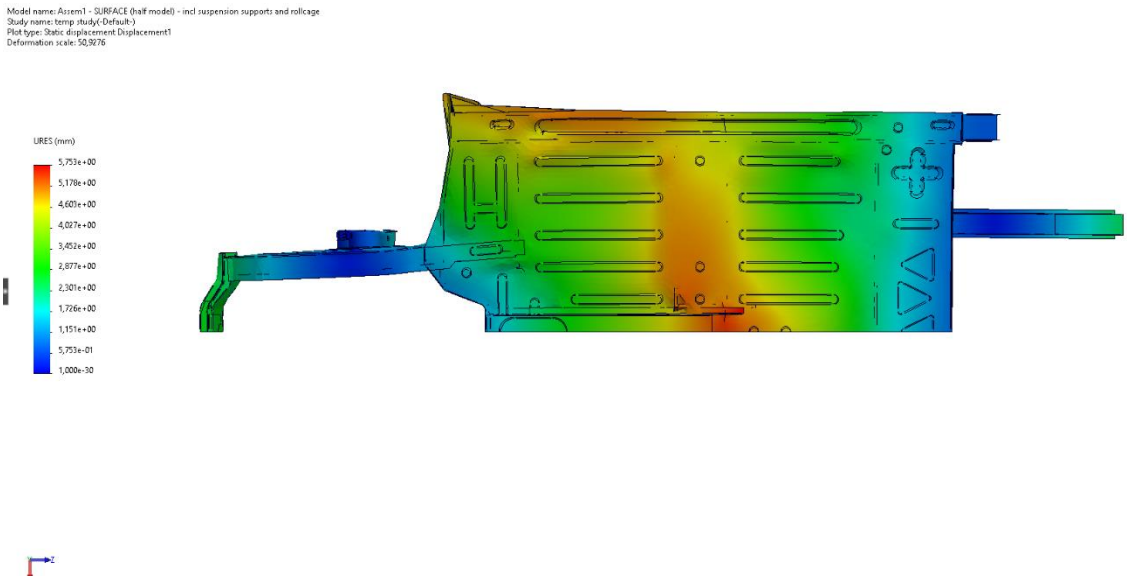


Figure S40: Bottom view of deformed chassis. (scaling factor ±50x)

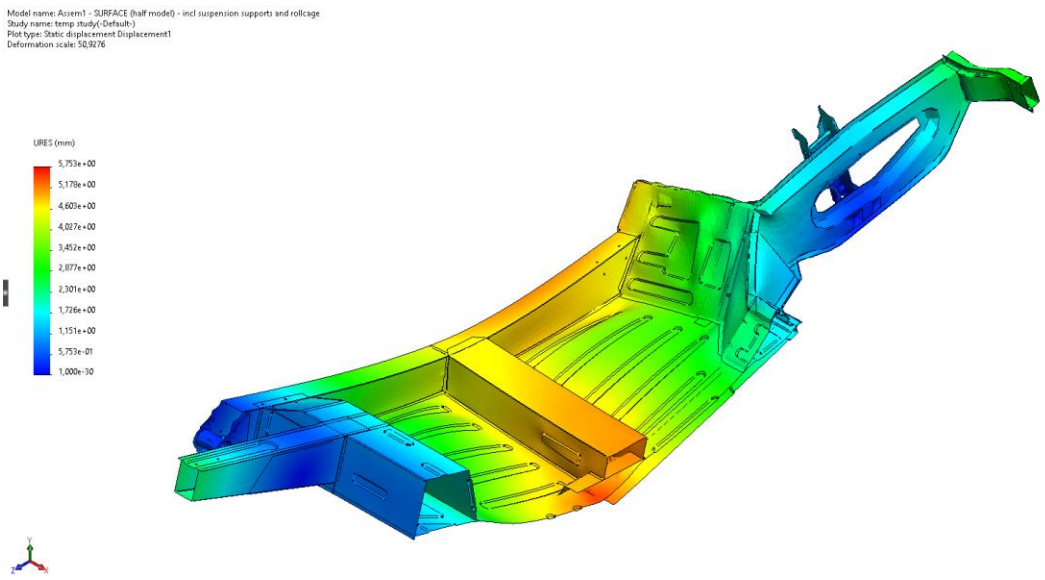


Figure S41: Isometric rear view (right side) of deformed chassis. (scaling factor ±50x)

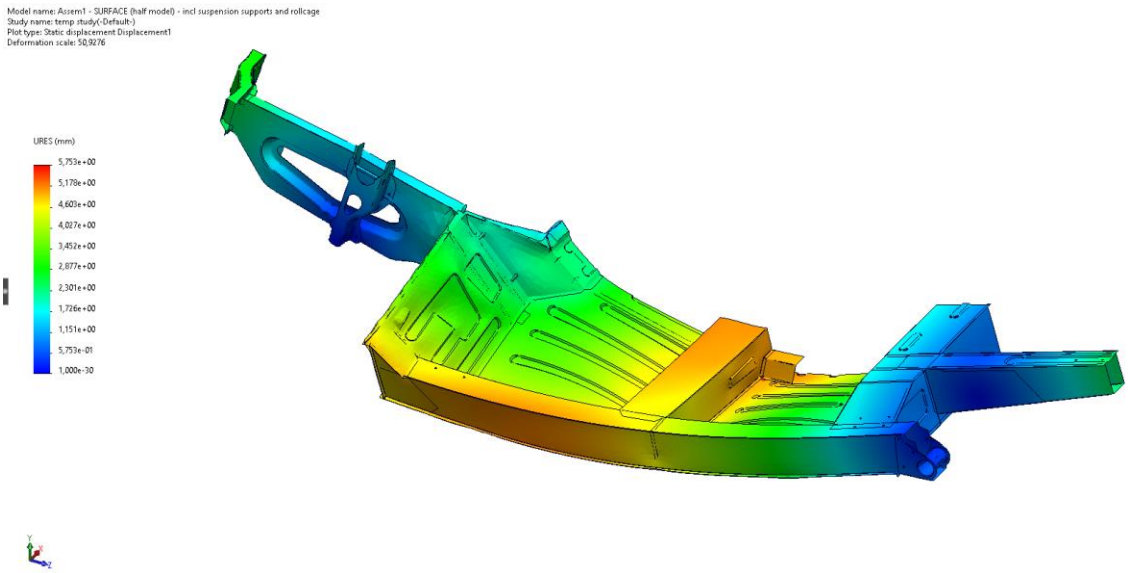
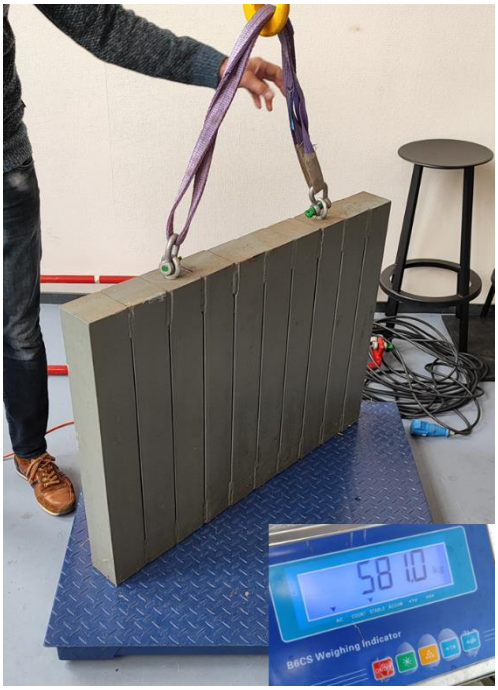
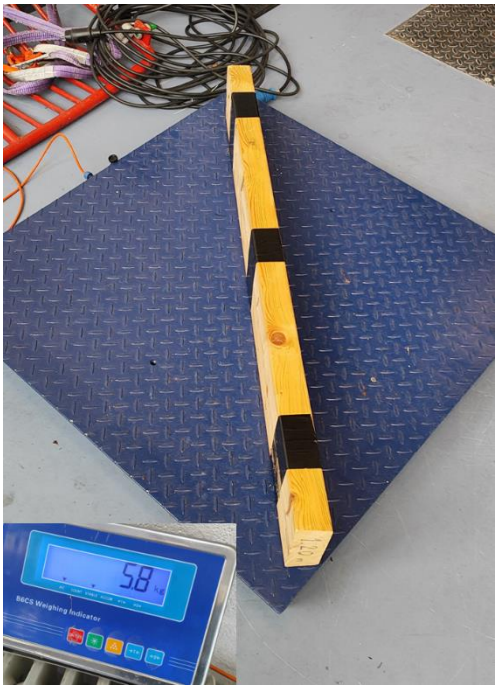


Figure S42: Isometric rear view (left side) of deformed chassis. (scaling factor ±50x)





Figures S43: The steel block weighing 581,0 kg and the wooden beam weighing 5,8 kg.



Figure S44: Dial gauges used to measure the deflection (made in the former USSR).

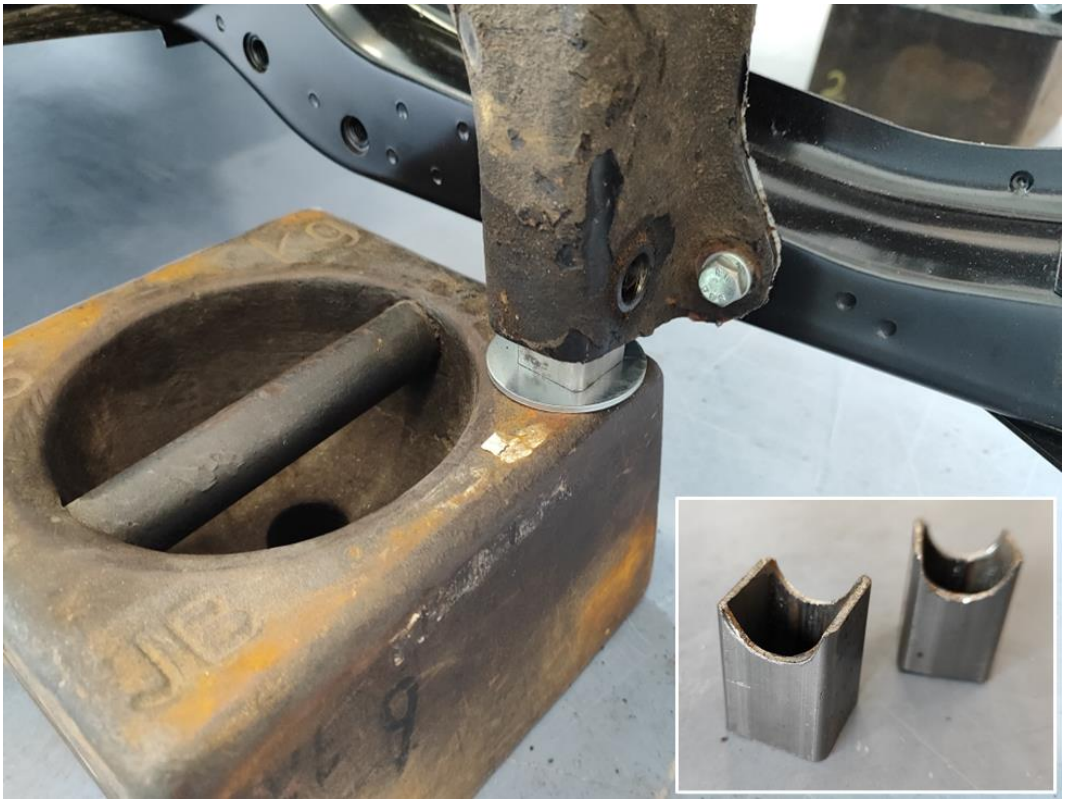


Figure S45: Support front section of the chassis.



Figure S46: Support rear section of the chassis.

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